Fakultas Informatika
School of Computing
Telkom University

Aplikasi Berbasis Platform

Intro Platform App

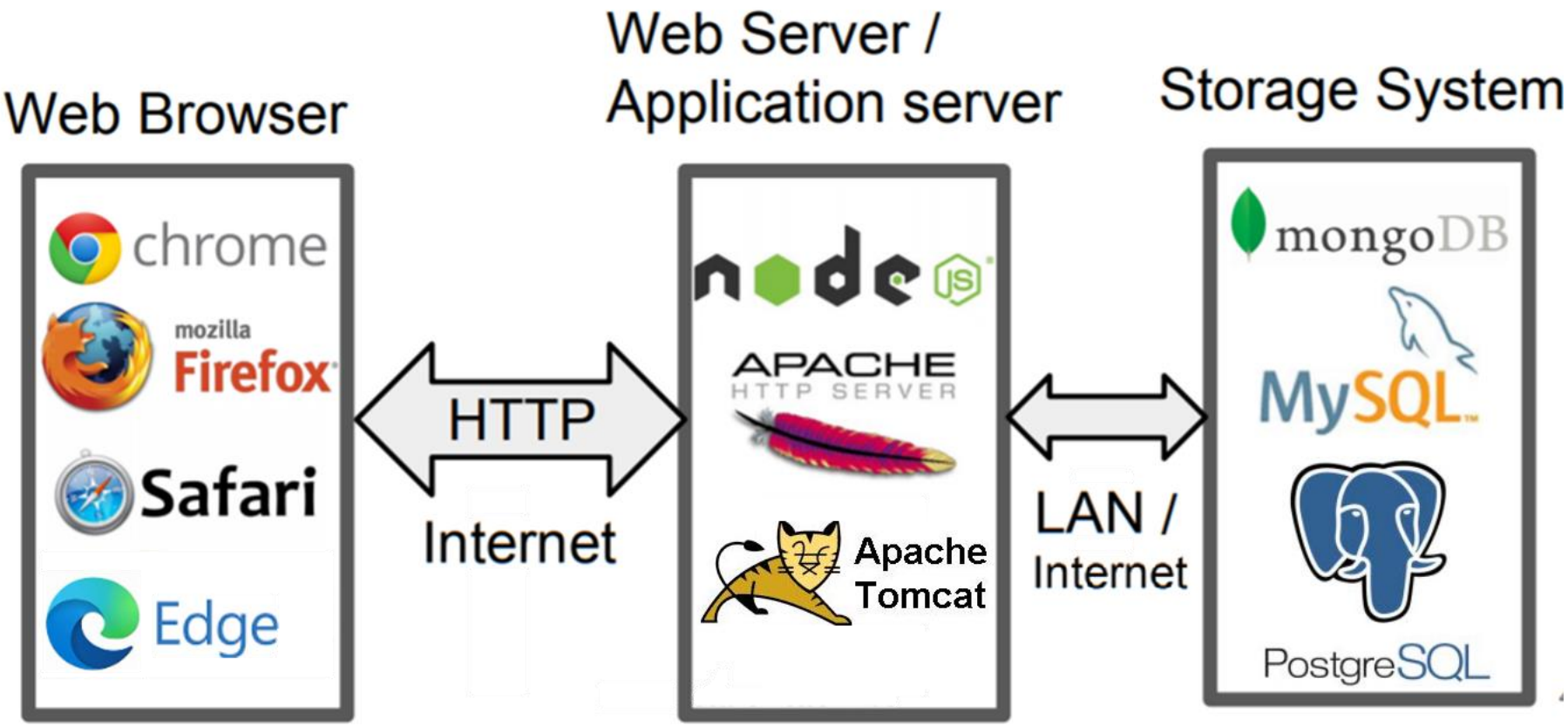
Outline

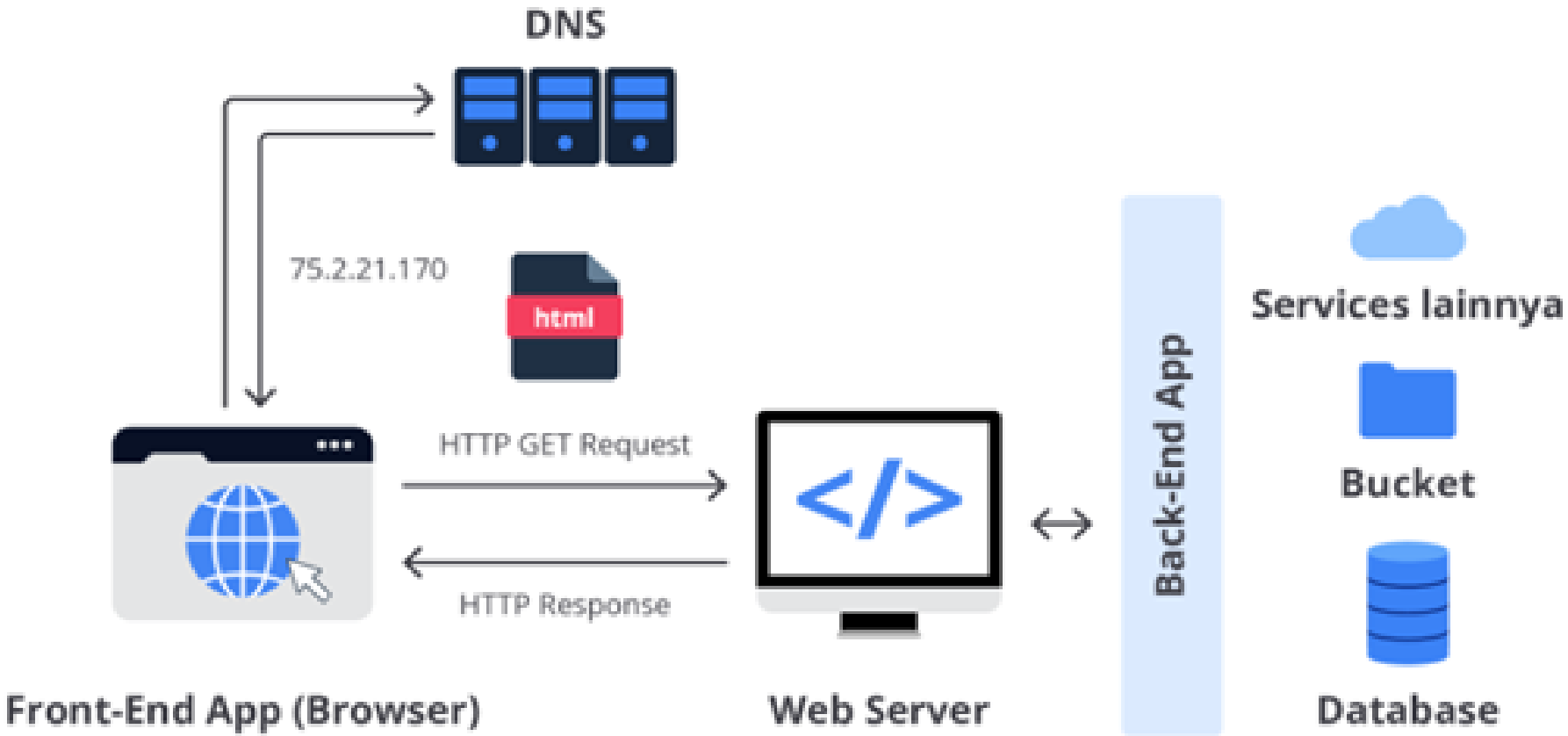
- Application Architecture
- Application Technologies
  - Web Frontend
- Tools

Platform App

- Native
  - Runs on PC
- Web
  - Runs on web browser
- Mobile
  - Runs on mobile device
  - Native, Hybrid, Native mobile web, PWA
- Embedded
  - Runs on IoT / specific hardware

Web Application Architecture




DNS
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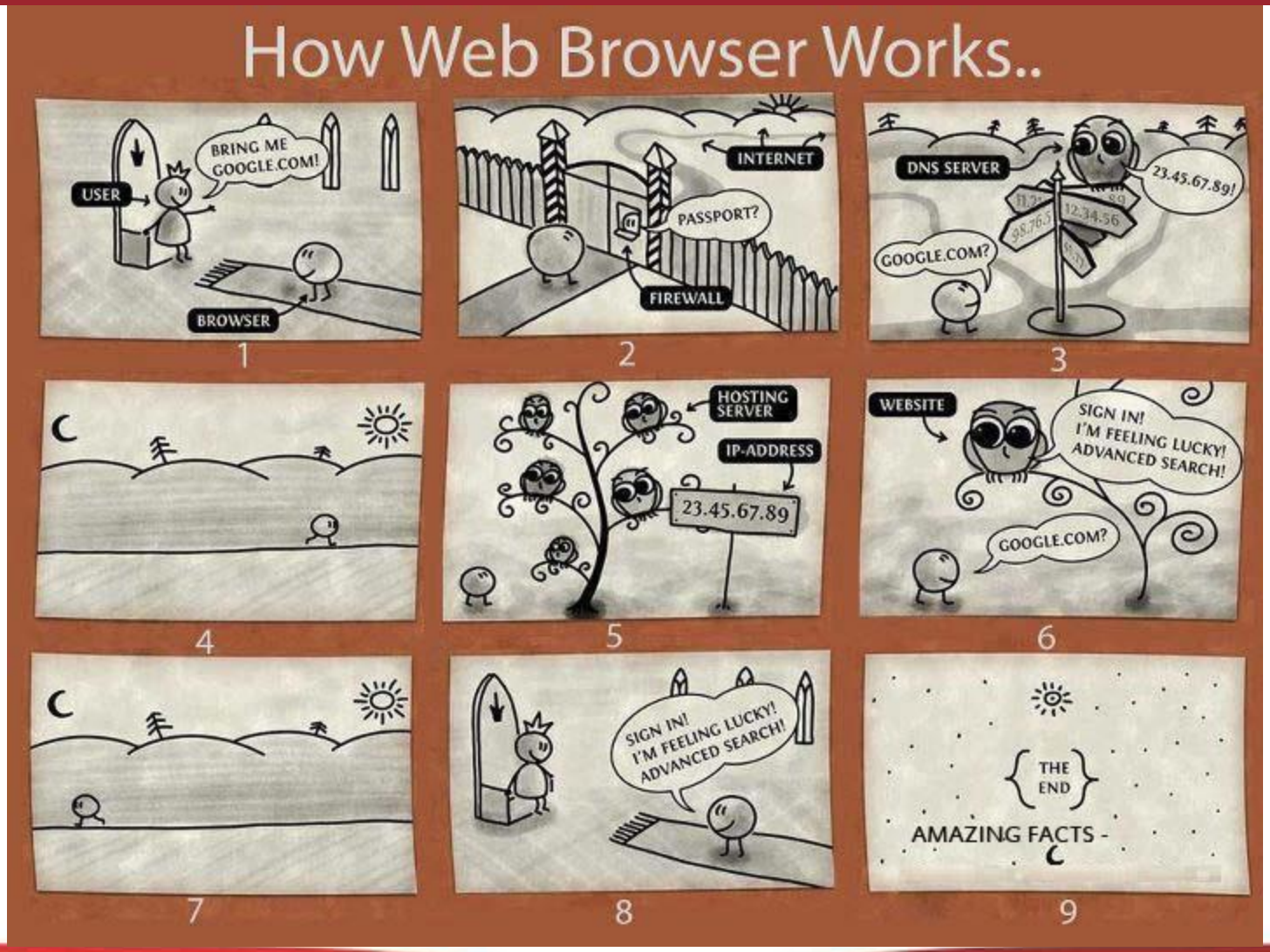
html
HTTP GET Request
HTTP Response
Front-End App (Browser)
Web Server
Back-End App
Services lainnya
Bucket
Database


How Web Browser Works..
USER
BRING ME GOOGLE.COM!
BROWSER
1
INTERNET
PASSPORT?
FIREWALL
2
DNS SERVER
23.45.67.89!
GOOGLE.COM?
3
4
HOSTING SERVER
IP-ADDRESS
23.45.67.89
5
WEBSITE
SIGN IN!
I'M FEELING LUCKY!
ADVANCED SEARCH!
GOOGLE.COM?
6
7
SIGN IN!
I'M FEELING LUCKY!
ADVANCED SEARCH!
8
THE END
AMAZING FACTS -
9

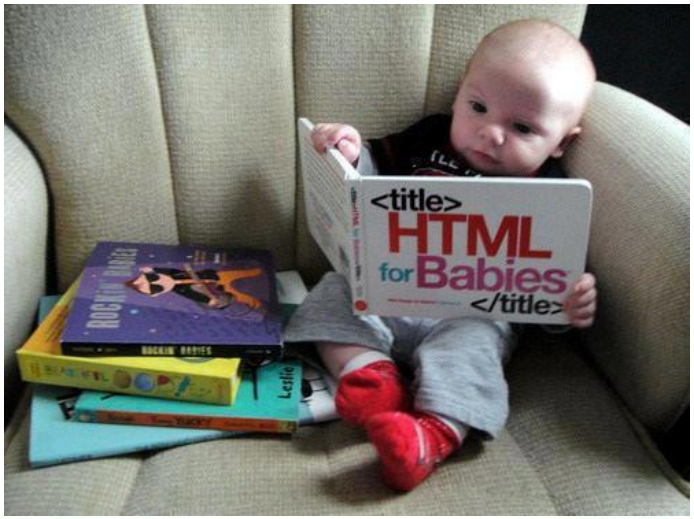
Web Technologies

- Client Side Scripting (Web Front-end)
  - HTML
  - CSS
  - Javascript

- Server Side Scripting (will be discussed in Week 3)

HTML

- Describe content and defines the structure within webpage.
- Hypertext documents on the web are designed to be linked to other documents.



HTML

- Markup Language: set of tags that are added to the text of web page to define its structure

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>Main Headline</h1>
<p>A paragraph.</p>

</body>
</html>
```

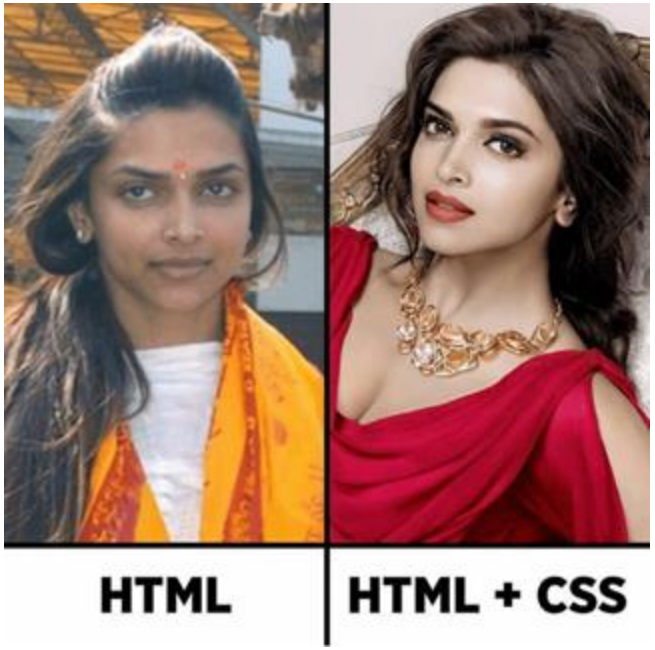
CSS

- Design plays an important part and is a key component of every successful website.
- Some people interested into finding the look and feel of a site, and enjoy working on the User Experience/UX.
- Some people prefer to work on the functionality of a site called User Interface/UI.
- All developers need to learn how CSS influences a document.

CSS

- Set of rules, or style sheets, for how the style is translated into visual form
- Cascading style rules priority of how the styles are rendered on a page

CSS



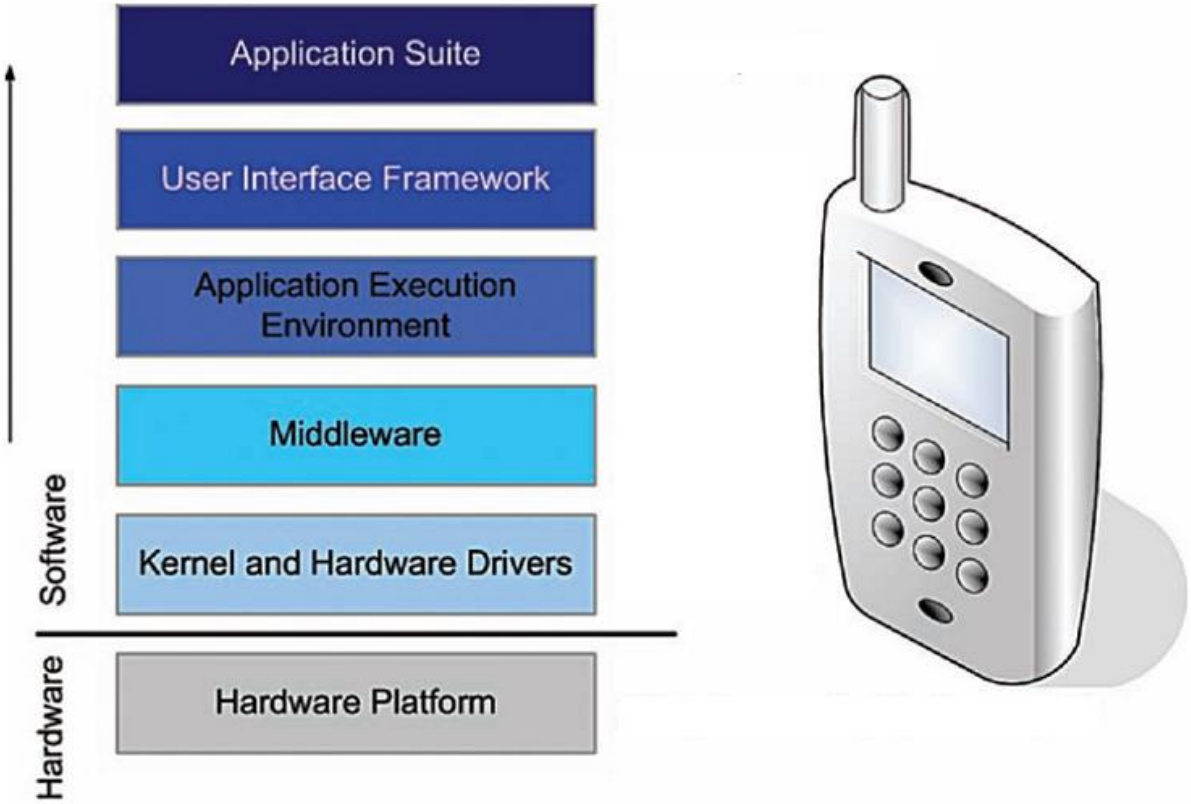
CSS

- Ultimate goal: Design one which compatible to all (Desktop, Mobile, Tablet)

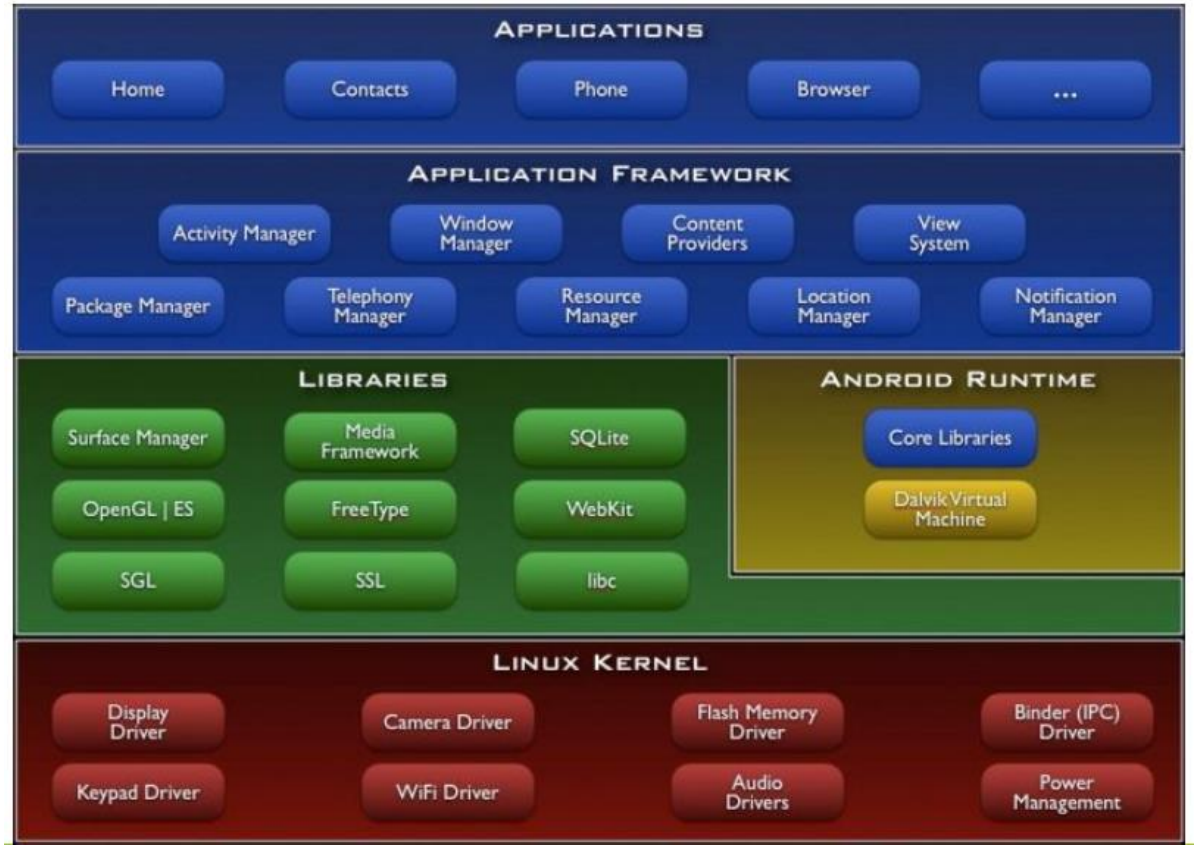
Javascript

- Language responsible for managing interaction within browser.
- Backend Developers using javascripts 50%
- Handling events, loading and unloading media, and creating modifying and deleting elements in a page
- Tools : jQuery, DataTables, Select2

Mobile Application Architecture



Android Application Architecture



Mobile Technologies

- Location-based Service
  - Maps, GPS
- Motion sensor
  - Accelerometer, Gyroscope, Camera
- Near Field Communication
- AI Camera
  - Image beautifying, Video stabilizer
- Payment Apps

Mobile App Development Challenges

- Small screen
- Various mobile device screen resolution and width
- Depend on internet connection
- Data Storage (full offline / full online / hybrid)

Tools Makes You Better

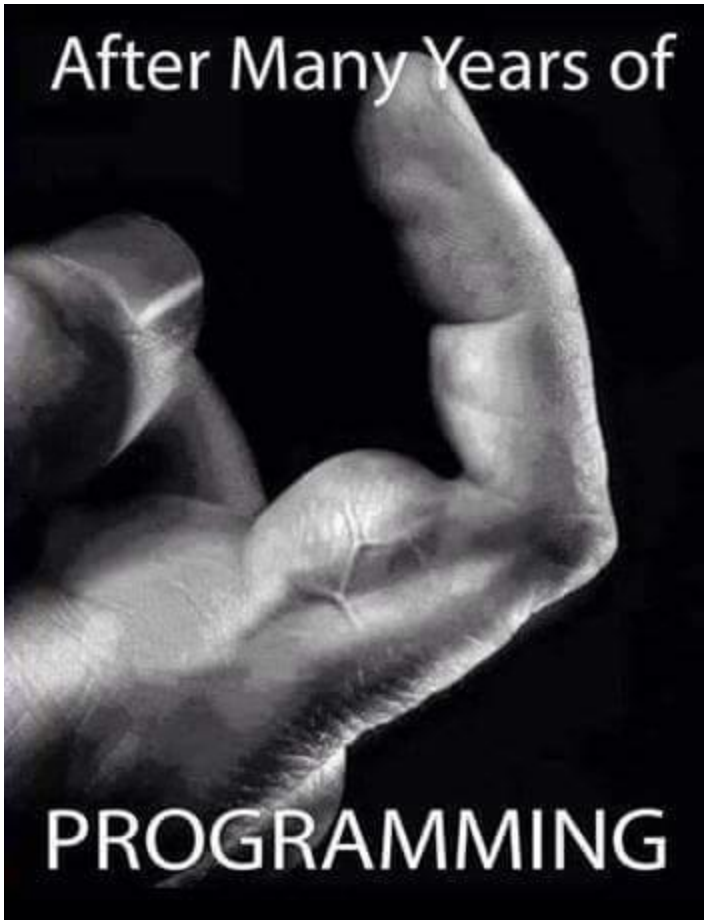
- IDE
  - Visual Studio Code (not Visual Studio 20xx)
  - Android Studio
- Emulator
  - If you don’t have Android phone
  - Warning: Uses lots of RAM, some hardwares / APIs don’t work properly

Web Developers Type

- Front-end Developer

Web Developers Type

- Back-end Developer (data & logical processing)
- Full-stack Developer (both frontend & backend)



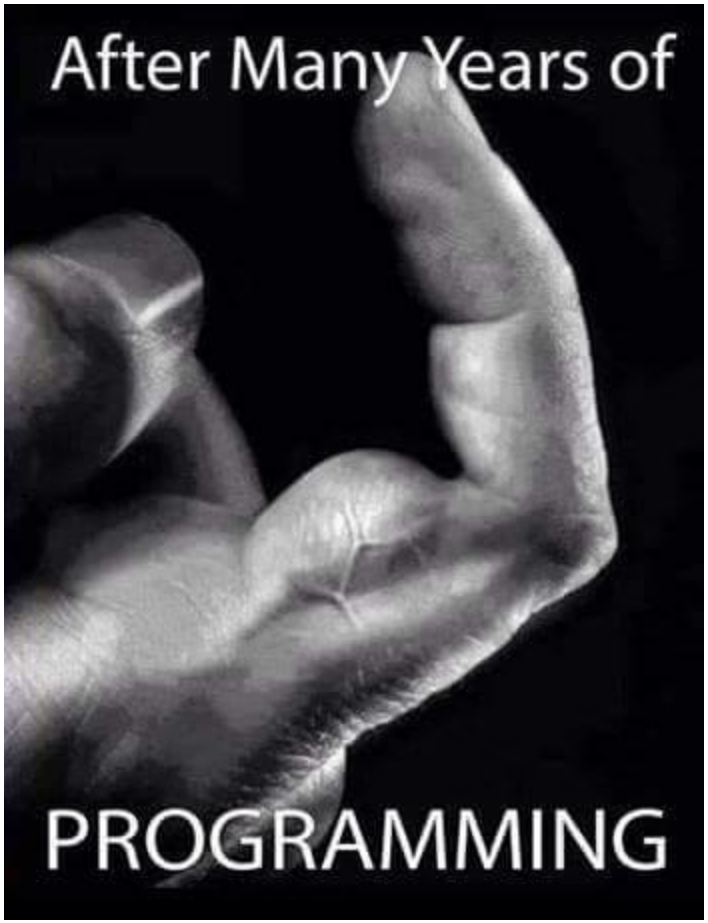
Mobile Developers Type

- Front-end Developer
  - Translate from design into layouts

A user interface is like a joke.
If you have to explain it,
it's not that good.

Mobile Developers Type

- Full-stack Developer
  - Translate from design into layouts
  - Create controller/model to bridge the frontend and backend/data

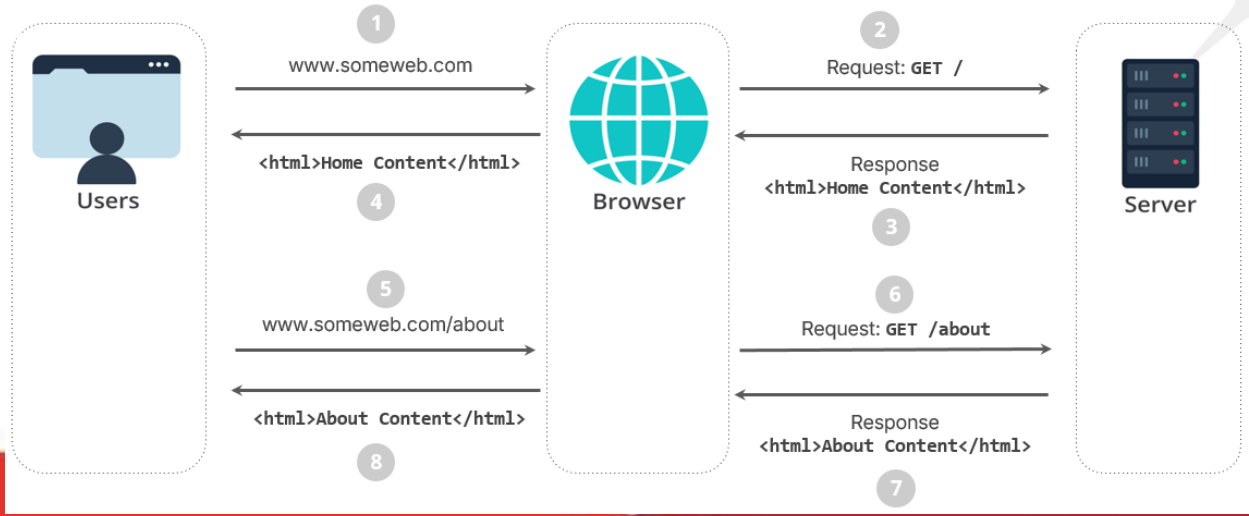


Front-end vs Back-end

- Front-end developers build everything that you see (user interface of web and apps).
- Back-end developers build the part that you don’t see (server-side of web and apps).

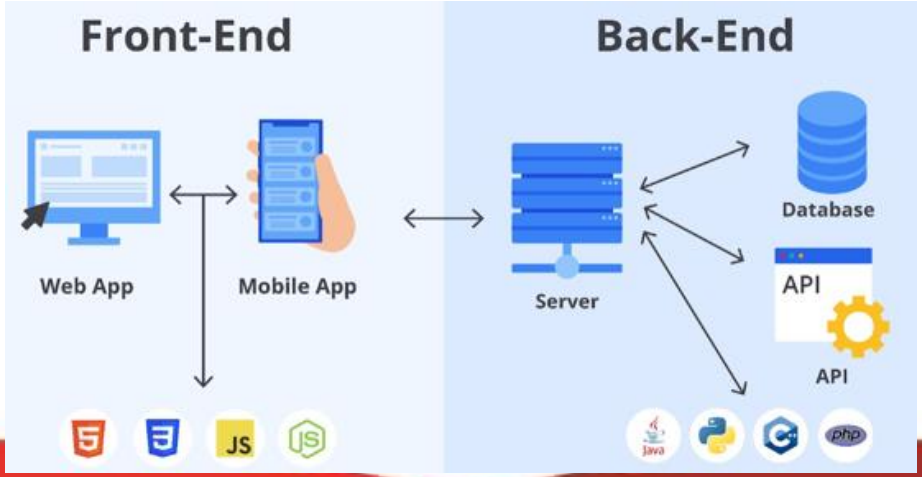
How to build Front-end and Back-end

- Server-Side Rendering
  - Backend loads a frontend file, fills it with data from backend (optional, if any), generates the file into a HTML page, and returns the page to browser
  - Usually used in monolithic development, where backend and frontend codes are in the same repository/codebase



How to build Front-end and Back-end

- Client-Side Rendering
  - Backend doesn't return a page, instead returns a formatted data to browser
  - Usually used in API-driven or microservices development (frontend and backend codes are in separate/different repository/codebase



Any question?

References

- Lynda.com : Web Development Foundations: Full-Stack vs Front-End
- Stanford : http://web.stanford.edu/class/cs142/lectures.html

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THANK YOU